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Github Link: https://github.com/Tanishq1234-alt/GEN_AI_Hands_on

Assignment 1: screenshot

```
... BERT Text Generation
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(
config.json: 100% ██████████ 570/570 [00:00<00:00, 44.0kB/s]
model.safetensors: 100% ██████████ 440M/440M [00:03<00:00, 217MB/s]
If you want to use `BertLMHeadModel` as a standalone, add `is_decoder=True.`
tokenizer_config.json: 100% ██████████ 48.0/48.0 [00:00<00:00, 2.72kB/s]
vocab.txt: 100% ██████████ 232k/232k [00:00<00:00, 3.26MB/s]
tokenizer.json: 100% ██████████ 466k/466k [00:00<00:00, 2.00MB/s]
Device set to use cpu
Truncation was not explicitly activated but `max_length` is provided a specific value, please use `truncation=True` to explicitly truncate ex
Both `max_new_tokens` (=256) and `max_length` (=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to the document
[{'generated_text': 'The future of Artificial Intelligence is.....'}
```

RoBERTa Text Generation

config.json: 100%  481/481 [00:00<00:00, 34.4kB/s]

model.safetensors: 100%  499M/499M [00:03<00:00, 222MB/s]

If you want to use 'RobertaLMHeadModel' as a standalone, add 'is_decoder=True.'

tokenizer_config.json: 100%  25.0/25.0 [00:00<00:00, 1.44kB/s]

vocab.json: 100%  899k/899k [00:00<00:00, 3.33MB/s]

merges.txt: 100%  456k/456k [00:00<00:00, 2.78MB/s]

tokenizer.json: 100%  1.36M/1.36M [00:00<00:00, 4.88MB/s]

Device set to use cpu

Truncation was not explicitly activated but 'max_length' is provided a specific value, please use 'truncation=True' to explicitly truncate ex

Both 'max_new_tokens' (=256) and 'max_length' (=50) seem to have been set. 'max_new_tokens' will take precedence. Please refer to the document

[{'generated_text': 'The future of Artificial Intelligence is'}]

```
> print("\nBART Text Generation")
```

```
bart_gen = pipeline("text-generation", model="facebook/bart-base")
```

```
output = bart_gen(prompt, max_length=50)
```

```
print(output)
```

BART Text Generation

config.json:  1.72k/? [00:00<00:00, 105kB/s]

model.safetensors: 100%  558M/558M [00:04<00:00, 245MB/s]

Some weights of BartForCausalLM were not initialized from the model checkpoint at facebook/bart-base and are newly initialized: ['lm_head.wei
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

vocab.json:  899k/? [00:00<00:00, 29.8MB/s]

merges.txt:  456k/? [00:00<00:00, 21.7MB/s]

tokenizer.json:  1.36M/? [00:00<00:00, 22.8MB/s]

Device set to use cpu

Truncation was not explicitly activated but 'max_length' is provided a specific value, please use 'truncation=True' to explicitly truncate ex

Both 'max_new_tokens' (=256) and 'max_length' (=50) seem to have been set. 'max_new_tokens' will take precedence. Please refer to the document

[{'generated_text': 'The future of Artificial Intelligence is raining raining raining ped bends bends bendsrecommUXAA EDT present raining rai

Some weights of the model checkpoint at bert-base-uncased were not used when initializing BertForMaskedLM: ['bert.pooler.dense.bias', 'bert.p
- This IS expected if you are initializing BertForMaskedLM from the checkpoint of a model trained on another task or with another architectur
- This IS NOT expected if you are initializing BertForMaskedLM from the checkpoint of a model that you expect to be exactly identical (initia
BERT Fill-Mask
Device set to use cpu
create : 0.5396932363510132
generate : 0.15575720369815826
produce : 0.05405500903725624
develop : 0.04451530799269676
add : 0.01757744885981083

Some weights of the model checkpoint at bert-base-uncased were not used when initializing BertForMaskedLM: ['bert.pooler.dense.bias', 'bert.pooler.dense.weight']

- This IS expected if you are initializing BertForMaskedLM from the checkpoint of a model trained on another task or with another architecture
- This IS NOT expected if you are initializing BertForMaskedLM from the checkpoint of a model that you expect to be exactly identical (initialization)

BERT Fill-Mask
Device set to use cpu
create : 0.5396932363510132
generate : 0.15575720369815826
produce : 0.05405500903725624
develop : 0.04451530799269676
add : 0.01757744885981083

```
print("\nRoBERTa Fill-Mask")

# RoBERTa uses <mask> instead of [MASK]
roberta_sentence = "The goal of Generative AI is to <mask> new content."

roberta_fill = pipeline("fill-mask", model="roberta-base")
roberta_results = roberta_fill(roberta_sentence)

for res in roberta_results:
    print(res["token_str"], ":", res["score"])
```

...

RoBERTa Fill-Mask
Device set to use cpu
generate : 0.3711312413215637
create : 0.3677145540714264
discover : 0.08351420611143112
find : 0.021335121244192123
provide : 0.016521666198968887

```
bart_sentence = "The goal of Generative AI is to <mask> new content."

print("\nBART Fill-Mask")
bart_fill = pipeline("fill-mask", model="facebook/bart-base")
bart_results = bart_fill(bart_sentence)

for res in bart_results:
    print(res["token_str"], ":", res["score"])
```

BART Fill-Mask
Device set to use cpu
create : 0.07461541891098022
help : 0.06571870297193527
provide : 0.060880109667778015
enable : 0.03593561053276062
improve : 0.03319477662444115

```
print("BERT Question Answering")

bert_qa = pipeline("question-answering", model="bert-base-uncased")
result = bert_qa(question=question, context=context)
print(result)
```

... Some weights of BertForQuestionAnswering were not initialized from the model checkpoint at bert-base-uncased and are newly initialized: ['qa_model.pooler.dense.bias', 'qa_model.pooler.dense.weight']

You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

BERT Question Answering
Device set to use cpu
{'score': 0.012080070562660694, 'start': 46, 'end': 67, 'answer': 'hallucinations, bias,'}

```
print("\nRoBERTa Question Answering")
```

```
roberta_qa = pipeline("question-answering", model="roberta-base")  
result = roberta_qa(question=question, context=context)  
print(result)
```

Some weights of RobertaForQuestionAnswering were not initialized from the model checkpoint at roberta-base and are newly initialized: ['qa_ou
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

RoBERTa Question Answering

Device set to use cpu

{'score': 0.0043637617491185665, 'start': 32, 'end': 82, 'answer': 'risks such as hallucinations, bias, and deepfakes.'}

```
print("\nBART Question Answering")
```

```
bart_qa = pipeline("question-answering", model="facebook/bart-base")  
result = bart_qa(question=question, context=context)  
print(result)
```

BART Question Answering

Some weights of BartForQuestionAnswering were not initialized from the model checkpoint at facebook/bart-base and are newly initialized: ['qa
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.

Device set to use cpu

{'score': 0.03568989131599665, 'start': 14, 'end': 81, 'answer': 'poses significant risks such as hallucinations, bias, and deepfakes'}

Assignment 2: Screenshots

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:  
The secret 'HF_TOKEN' does not exist in your Colab secrets.  
... To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in  
You will be able to reuse this secret in all of your notebooks.  
Please note that authentication is recommended but still optional to access public models or datasets.  
warnings.warn(  
tokenizer_config.json: 100% ██████████ 26.0/26.0 [00:00<00:00, 549B/s]  
vocab.json: 100% ██████████ 1.04M/1.04M [00:00<00:00, 16.4MB/s]  
merges.txt: 100% ██████████ 456k/456k [00:00<00:00, 7.73MB/s]  
tokenizer.json: 100% ██████████ 1.36M/1.36M [00:00<00:00, 16.9MB/s]  
config.json: 100% ██████████ 665/665 [00:00<00:00, 19.3kB/s]  
model.safetensors: 100% ██████████ 548M/548M [00:06<00:00, 144MB/s]  
generation_config.json: 100% ██████████ 124/124 [00:00<00:00, 5.12kB/s]  
The attention mask is not set and cannot be inferred from input because pad token is same as eos token. As a consequence, you may observe une  
Prompt: The knight entered the dark cave  
  
Generated Story:  
  
The knight entered the dark cave, where a strange sound from outside of his skull was heard. He looked over to see who it belonged on top of  
Rising down upon Rangiku's back she suddenly felt her cheeks bulge open at hearing such sounds again while looking forward towards its direct
```